UM-2 analysis

The x_0 mass mechanism, and the validity of associating dark matter and dark energy with the dirac16complex field

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Analysis, version 3, 2026-10-18 (version 2: 2026-10-18; version 1: 2026-10-17)

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1 How to read this document

The author asked for two analyses: of the mechanism that gives every non-massless particle its mass through the dependence of its wave function on the hidden space coordinate x_0 (periodic with period $\pi/(6H)$, the uncertainty relation $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$ being the physical reason), and of the validity of associating dark matter and dark energy with dirac16complex. This document carries out both within UM-2. It is self-contained: every equation it uses is written here. Labels: AUTHOR, RECORD, STANDARD, ASSUMED, INVENTED, DERIVED (computed and verified), ESTIMATE, FROM MEMORY (a measured bound quoted from memory), OPEN. Units $\hbar = c = 1$. Coordinates are the author's $x_0, \ldots, x_7$ (x_0 hidden space, x_1, x_2, x_3 space, x_4 time, x_5, x_6, x_7 extra times); W^+ has charge $+e$, W^- charge $-e$.

Version 2 (2026-10-18) analyses the model with three generations of quarks and leptons and three neutrino flavours with their mixing: twelve fermion masses instead of four, the neutrino masses and mixing as x_0 observables, the dark matter of both halves of dirac16complex (the chiral tip Z portal of version 1 is removed), and fold bands that are never thermalised.

Version 3 (2026-10-18) analyses the completed solution: the dark matter with the new portal I13, which gives it the observed relic abundance within the unitarity limit (and couples to the baryon-number current, because a $B-L$ current would thermalise the right-handed neutrinos); the fold bands produced by freeze-in through I11 (radiation, $\Delta N_{eff} = 0.001$); the nucleon mass as the most block-sensitive mass, through the QCD scale; the Kaluza-Klein gravitons, which do not couple to the gluon energy; the Majorana alternative, which the x_0 mechanism disfavours; and the back-reactions of the coupled solution, which are all small.

Symbols with more than one use: $\xi = y/L$ is the scaled hidden coordinate (primes denote $d/d\xi$, except $G'(k) = dG/dk$); dots denote d/dx_4, and $h = \dot{f}/f$ is the rate of change of the block scale (h_ν, with a subscript, the comoving heating of the neutrinos); $H = 14.296$ TeV is the eight-dimensional scale and H_0 the Hubble constant; $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue of an x_0 operator, while in the formula for the four-dimensional cosmological term Λ_4 the symbol $\Lambda = -10H^2$ is the eight-dimensional cosmological constant; $\Lambda_\star$ is the scale of the effective interactions; k is the momentum of a mode along x_1, x_2, x_3 in h_j and in $\varepsilon = e^{-a_4}|k|$, and the block variable $1/f^2$ in k_*, $G(k)$ and Λ_4; ρ is an energy density (ρ_0 the electroweak parameter) and ϱ a probability density in ξ; w is an equation-of-state parameter (w_0, w_a of the form $w = w_0 + w_a(1-a)$, a the scale factor), except in the fermion mass condition ($w = \sqrt{c^2 - e^2}$) and the Lovelock weights w_2, w_3; $z = 6Hx_0$, except where it is a redshift. $s = fy$ is the proper hidden length (Δs its spread), except in $s = \sinh w/w$ and s_L, s_R of the fermion mass condition and $s_{ij} = \sin\theta_{ij}$ of the mixing matrices; p is the momentum conjugate to ξ ($\langle p^2\rangle$), the hidden wave number of the dark matter ($p\,fL = 1.899$) and a pressure (p_0, p_3, p_t, and p of the condensate); a is the scale factor ($a = e^{a_4}$), while a_L, a_R, b_L, b_R are spinor components at the tip; e is the eigenvalue εfL of a fermion, the electron and the elementary charge; μ is a bulk mass (μ_L, μ_Q, μ_R), μ_N the dark-matter-nucleon reduced mass and μ the muon; $i, j = 1, 2, 3$ label generations and neutrino mass states, $\alpha \in \{e, \mu, \tau\}$ flavours (α is also the fine-structure constant, α_s the strong coupling); U is the PMNS and V the CKM matrix; κ (κ_ν, κ_e, κ_u, κ_d, κ_f) a tip matrix (κ_M the Majorana tip coupling); G_V, G_A the strengths of I11 and I13, $\tilde{I}$, $\tilde{I}^{(b)}$ overlap integrals, $P_b = (1-\gamma^{x_0})/2$, J^B the baryon-number current, $\Lambda^{(n_f)}$ the QCD scale with n_f flavours, $\sigma_{\pi N}$, σ_s the nucleon sigma terms; $G(k)$ is the gravity factor of the block, G_N Newton's constant and G_g, G_W, G_B the four-dimensional couplings of the dark matter to gluons, W and B fields (G without a subscript their combination, G^α the gluon field); F is a field strength, F_h the decoupling factor of ΔN_{eff}; g is the degeneracy of a momentum shell in the Kohn-Sham sums or the number of internal states of a particle, g_* and g_s the effective numbers of relativistic degrees of freedom (energy, entropy), g_i the gauge couplings.

Summary of the verdicts.

question	verdict
Can every mass come from the x_0 dependence of wave functions periodic with period $\pi/(6H)$?	YES (DERIVED): every four-dimensional mass of UM-2 ($W^\pm$, Z, the twelve fermions of three generations, the dark matter, the Kaluza-Klein modes) is an eigenvalue of an x_0 operator; graviton, photon, gluons and the dirac16complex fold bands are exactly massless
Is the uncertainty relation the reason for the masses?	ONLY FOR THE HEAVY STATES: the uncertainty estimate lies below the masses of the computed Kaluza-Klein and odd excitations (and of the dark matter), but the light masses ($W^\pm$, Z and all twelve fermions) lie 3×10^{-3} (top quark) to 10^{-17} (lightest neutrino) below the uncertainty estimate and come from the tip of the x_0 cell
Are the neutrino masses and their mixing x_0 effects?	YES for the masses, NO for the mixing angles: the three neutrino masses are x_0 eigenvalues (the most sensitive to the block of all masses), the PMNS matrix is the tip matrix κ_ν, an input; given both, the flavour conversion follows

question	verdict
Do the extra times deflate to irrelevance?	IN THE PRIMORDIAL ERA, then FROZEN: they deflate by up to a factor 2.9 into the stabilised block; a continuing deflation is excluded because every mass and coupling depends on the block size
Is dirac16complex dark matter?	YES, CONDITIONALLY (version 3): the lightest odd quanta of its two halves Ψ_A and Ψ_B (two degenerate Dirac fermions of 44.35 TeV) are cold, neutral and stable (DERIVED, conditional); the gauge-kinetic portal I12 alone cannot produce their relic abundance within perturbative unitarity (version 2), but with the s-wave portal I13 (INVENTED) they freeze out with $\Omega h^2 = 0.060$ each within the unitarity limit, and their scattering on xenon is 0.16 of the present LZ limit (testable); the massless fold bands are radiation ($\Delta N_{eff} = 0.001$ by freeze-in), not dark matter
Is dirac16complex dark energy?	NOT SUPPORTED: no positive-energy state has $w < 0$; a negative-coupling condensate runs away; the dark energy of UM-2 is a tuned four-dimensional cosmological term
Does the x_0 mechanism fix the nucleon mass?	INDIRECTLY (version 3): the nucleon mass is QCD binding energy, $\propto \Lambda_{QCD}$, set by the gauge coupling at the Kaluza-Klein scale, itself an x_0 quantity; $d\ln m_p/d\ln f = -46$ makes it the most block-sensitive mass of UM-2

2 The x_0 mass mechanism

2.1 The author's requirement in equations

The hidden coordinate enters through $z = 6Hx_0$; one period is $\ell_0 = \pi/(6H)$; on each half of a period

$$y = \frac{\ln|\sin z|}{6H} \in (-L, 0], \qquad dy = \cot z\, dx_0, \qquad ds_{hidden} = f\, dy,$$

the fold $y = 0$ ($z = \pi/2$) joining the two halves and the tip $y = -L$ ($\eta = HL = 3$) closing them. Every wave function of UM-2 depends on x_0 through $y(z)$, so it is periodic in x_0 with period $\pi/(6H)$. The uncertainty relation, in the proper hidden length $s = fy$, reads $\Delta s\, \Delta p_s \geq \frac{1}{2}$ on an unbounded line (on the cell, which ends at the tips, it acquires a tip term; below), and the author's estimate of a mass is

$$m_H = \frac{\hbar c}{2\Delta s}, \qquad \Delta s = fL\, \Delta\xi, \qquad \Delta\xi^2 = \frac{\int_{-1}^{0} \xi^2 \varrho(\xi)\, d\xi}{\int_{-1}^{0} \varrho(\xi)\, d\xi},$$

the root-mean-square proper distance from the fold of the probability density ϱ in the flat measure ($\varrho = \tilde{u}^2/\int \tilde{u}^2$, $\tilde{u} = e^{3\eta\xi/2}u$, for bosons; $\varrho = |\chi|^2/\int|\chi|^2$ for fermions; integrals over $[-1, 0]$), $\xi = y/L$; over the full period the density is symmetric about the fold, so $\Delta\xi$ is also the spread of the full-period profile.

2.2 Every mass is an x_0 eigenvalue (DERIVED)

With $\xi = y/L$:

- **Gauge bosons and the graviton**: $-e^{-3\eta\xi}(e^{3\eta\xi}u')' = \Lambda u$, $\Lambda = (m\, fL)^2$; fold $u'(0) = 0$ (even) or $u(0) = 0$ (odd); tip $u'(-1) = \beta u(-1)$ with $\beta = 0$ for the photon, the gluons and the graviton and $\beta_W = 0.011912$ (W^+, W^-), $\beta_Z = \beta_W/\cos^2\theta_W$ (Z; θ_W the angle of the gauge couplings, $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$, fixed by the measured masses and ρ_0).

- **Fermions** (three generations): $h_j = j[-\frac{i}{f}\sigma_1\partial_y + \mu\sigma_2 + e^{-a_4}k\sigma_3]$ (k the momentum along x_1) with the chiral boundary conditions. With $c = f\mu L$, $e = \varepsilon f L$ and $k = 0$, the components (a, b) of a block $j = +1$ obey

$$a' = c\,a - e\,b, \qquad b' = e\,a - c\,b;$$

the fold condition is $b(0) = 0$ for a doublet ($c_L = f\mu_L L > 0$) and $a(0) = 0$ for a singlet ($c_R = f\mu_R L < 0$), whose solutions $(a_L, b_L) = (\cosh w\xi + (c_L/w)\sinh w\xi,\ (e/w)\sinh w\xi)$ and $(a_R, b_R) = (-(e/w)\sinh w\xi,\ \cosh w\xi - (c_R/w)\sinh w\xi)$ reach the tip $\xi = -1$ with $(a_L, b_L) = (C_L - c_L s_L, -s_L e)$ and $(a_R, b_R) = (s_R e, C_R + c_R s_R)$, $s = \sinh w/w$, $C = \cosh w$, $w = \sqrt{c^2 - e^2}$. The tip conditions $b_L = \kappa b_R$, $a_R = -\kappa^\dagger a_L$ (vectors over the generations, κ the 3×3 tip matrix of a sector) give

$$\Big[\mathrm{diag}(b_{L\alpha}) + \kappa\,\mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\kappa^\dagger\,\mathrm{diag}(a_{L\alpha})\Big]\mathcal{A} = 0;$$

with one doublet bulk mass per sector and a unitary κ the determinant factorises into $\prod_i (b_L a_{Ri} + a_L b_{Ri})/a_{Ri}$, so each mass is the root of the one-generation condition

$$D(e) = e^2 s_L s_R - (C_L - c_L s_L)(C_R + c_R s_R) = 0, \qquad e \simeq \zeta(c_L)\zeta(-c_R), \quad \zeta(c)^2 = \frac{c\,e^{-c}}{\sinh c},$$

and the left-handed mixing matrix of the sector is κ itself ($\kappa_\nu = U_{PMNS}$, $\kappa_e = 1$, $\kappa_u = 1$, $\kappa_d = V_{CKM}$; the roots of the 3×3 determinant agree with the one-generation roots, both refined in the same 50-digit computation, to 3.0×10^{-42}; the stored masses agree with them to 4.5×10^{-11}).
- **dirac16complex**: h_j with $M = m$, fold $\chi_2(0) = 0$ (even) or $\chi_1(0) = 0$ (odd), tip $\chi_2(-L) = 0$.

The masses (DERIVED): photon, gluons, graviton 0 exactly (flat profile); dirac16complex fold bands 0 exactly (chiral zero modes, $\varepsilon = e^{-a_4}|k|$); $W^\pm$ 80.36 GeV and Z 91.19 GeV (both reproduced: β_W fitted to ρ_0, the angle of the couplings to m_W/m_Z); the twelve fermions with the doublet localisations $c_L = 5.9529$ (leptons) and $c_Q = 3.3823$ (quarks) and the singlet localisations

fermion	ν_1	ν_2	ν_3	e	μ	τ
mass	1.000 meV	8.712 meV	50.140 meV	0.511 MeV	105.66 MeV	1776.93 MeV
c_R	−35.0452	−32.8481	−31.0702	−14.5539	−8.9809	−5.9529

fermion	u	c	t	d	s	b
mass	2.16 MeV	1.273 GeV	172.56 GeV	4.70 MeV	93.5 MeV	4.183 GeV
c_R	−15.4300	−8.7683	−3.3823	−14.6258	−11.5158	−7.5005

(the heaviest member of each sector has $c_R = -c_L$; m_1 is ASSUMED, m_2, m_3 follow from the measured splittings); the dark matter $3.102204\,m = 44.35$ TeV; Kaluza-Klein levels from 121 TeV; every odd mode and every excitation of the gauge fields and the graviton above the gap $3H/(2f_*) = 99.5$ TeV; the Ψ excitations from its lowest odd level, 44.35 TeV, upwards.

2.3 What the uncertainty relation does and does not bound (DERIVED)

For a boson, write $u = e^{-3\eta\xi/2}\tilde{u}$. The exact identity

$$e^{3\eta\xi}u'^2 = \tilde{u}'^2 + \frac{9}{4}\eta^2\tilde{u}^2 - \frac{3}{2}\eta\,(\tilde{u}^2)'$$

turns the Rayleigh quotient $(m\,fL)^2 = \big[\int e^{3\eta\xi}u'^2 d\xi + \beta e^{-3\eta}u(-1)^2\big]/\int e^{3\eta\xi}u^2 d\xi$ into the decomposition

$$(m\,fL)^2 = \langle p^2\rangle + \frac{9\eta^2}{4} - \frac{3\eta}{2}\varrho(\text{fold}) + \Big(\frac{3\eta}{2} + \beta\Big)\varrho(\text{tip}), \qquad \langle p^2\rangle = \frac{\int \tilde{u}'^2}{\int \tilde{u}^2}, \quad \varrho = \frac{\tilde{u}^2}{\int \tilde{u}^2},$$

and for a fermion $(\varepsilon f L)^2 = \langle|\chi'|^2\rangle + c^2 - c\,[\chi^\dagger\sigma_3\chi]_{-1}^0/\int|\chi|^2$, with $\langle p^2\rangle = \langle|\chi'|^2\rangle$ and $\varrho = |\chi|^2/\int|\chi|^2$ (integrals over $\xi \in [-1, 0]$). Only the hidden kinetic term is bounded by the uncertainty relation. On the cell, which ends at the tips, the relation reads

$$\sqrt{\langle p^2\rangle}\,\Delta\xi \geq \frac{1 - \varrho(\text{tip})}{2},$$

because $1 = \int_{-1}^0 \varrho\, d\xi = \varrho(\text{tip}) - \int_{-1}^0 \xi\varrho'\, d\xi$ (integration by parts; the fold end $\xi = 0$ contributes nothing) and the Cauchy-Schwarz inequality gives $|\int \xi\varrho'\, d\xi| \leq 2\Delta\xi\sqrt{\langle p^2\rangle}$. The familiar form $\langle p^2\rangle \geq 1/(4\Delta\xi^2)$ is guaranteed only for profiles negligible at the tip (it can hold otherwise: the first even Kaluza-Klein mode, $\varrho(\text{tip}) = 0.655$, has $\langle p^2\rangle\Delta\xi^2 = 4.59$); the broad fold zero mode of Ψ violates it ($\langle p^2\rangle\Delta\xi^2 = 0.097$, $\varrho(\text{tip}) = 0.489$) and satisfies the relation with the tip term. Computed for every class of particle (DERIVED; the sum rules reproduce the boson eigenvalues $(m\,fL)^2$ to 1.0×10^{-8} in absolute value, the discretisation error of the finite-element profiles — relative 7.8×10^{-4} for $W^\pm$ and 6.0×10^{-4} for Z, whose eigenvalues are only 1.3 and 1.7×10^{-5}, and below 3×10^{-11} for the excitations — and the dark matter to 1.2×10^{-11} relative; the relation with the tip term, a theorem, holds for every tabulated state, with the smallest margin 0.057 for the Ψ fold band):

Bosons (the warp term is $9\eta^2/4 = 20.25$ for every mode):

state	$\langle p^2\rangle$	fold term	tip term	$(m\,fL)^2$	m/m_H
photon, gluons, graviton	20.250	−40.505	0.0050	0	0
$W^\pm$	20.250	−40.505	0.0050	1.32×10^{-5}	1.14×10^{-3}
Z	20.250	−40.505	0.0050	1.70×10^{-5}	1.29×10^{-3}
first even Kaluza-Klein mode	9.870	−2.949	2.949	30.120	7.49
first odd mode	4.891	0	1.948	27.089	6.47

The chiral zero modes of the fermions: each one alone is massless, its kinetic term $\langle|\chi'|^2\rangle$ equals the bulk term c^2 and the boundary term is $-2c^2$; one doublet zero mode serves the three generations of a sector, and every fermion has its own singlet zero mode:

zero mode	c	$\langle|\chi'|^2\rangle = c^2$	boundary term	Δs (m)	m_H (GeV)	ϱ(tip)	$\langle p^2\rangle\Delta\xi^2$
lepton doublet	5.9529	35.437	−70.874	1.06×10^{-21}	93 080	8.0×10^{-5}	0.4997
quark doublet	3.3823	11.440	−22.880	1.83×10^{-21}	53 800	7.8×10^{-3}	0.4829
ν_1 singlet	35.0452	1228.16	−2456.33	1.80×10^{-22}	547 816	2.5×10^{-29}	0.5000
ν_2 singlet	32.8481	1079.00	−2157.99	1.92×10^{-22}	513 472	1.9×10^{-27}	0.5000
ν_3 singlet	31.0702	965.36	−1930.71	2.03×10^{-22}	485 680	6.4×10^{-26}	0.5000
e singlet	14.5539	211.82	−423.63	4.34×10^{-22}	227 503	6.6×10^{-12}	0.5000
μ singlet	8.9809	80.657	−161.314	7.03×10^{-22}	140 388	2.8×10^{-7}	0.5000
τ singlet	5.9529	35.437	−70.874	1.06×10^{-21}	93 080	8.0×10^{-5}	0.4997
u singlet	15.4300	238.09	−476.17	4.09×10^{-22}	241 198	1.2×10^{-12}	0.5000
c singlet	8.7683	76.883	−153.766	7.20×10^{-22}	137 064	4.2×10^{-7}	0.5000
t singlet	3.3823	11.440	−22.880	1.83×10^{-21}	53 800	7.8×10^{-3}	0.4829
d singlet	14.6258	213.91	−427.83	4.32×10^{-22}	228 626	5.8×10^{-12}	0.5000
s singlet	11.5158	132.61	−265.23	5.48×10^{-22}	180 012	2.3×10^{-9}	0.5000

zero mode	c	$\langle\lvert\chi'\rvert^2\rangle = c^2$	boundary term	Δs (m)	m_H (GeV)	ϱ(tip)	$\langle p^2\rangle\Delta\xi^2$
b singlet	7.5005	56.258	-112.516	8.41×10^{-22}	117 248	4.6×10^{-6}	0.5000

(A singlet zero mode is right-handed with $c_R<0$; its profile is that of a left-handed zero mode with $c=-c_R$. A zero mode localised exponentially at the fold with $\varrho\propto e^{2c\xi}$ has $\Delta\xi^2\simeq1/(2c^2)$ and so $\langle p^2\rangle\Delta\xi^2\simeq\frac{1}{2}$.) The fermions, their uncertainty estimates (that of the broader of their two zero modes, i.e. the smaller m_H) and their sensitivity to the block:

fermion	mass	m_H (GeV)	m/m_H	$d\ln m/d\ln c_R$	$d\ln m/d\ln f$
ν_1	1.000 meV	93 080	1.1×10^{-17}	-34.55	-41.0
ν_2	8.712 meV	93 080	9.4×10^{-17}	-32.35	-38.8
ν_3	50.140 meV	93 080	5.4×10^{-16}	-30.57	-37.0
e	0.511 MeV	93 080	5.5×10^{-9}	-14.05	-20.5
μ	105.66 MeV	93 080	1.1×10^{-6}	-8.48	-14.9
τ	1776.93 MeV	93 080	1.9×10^{-5}	-5.45	-11.9
u	2.16 MeV	53 800	4.0×10^{-8}	-14.93	-18.8
c	1.273 GeV	53 800	2.4×10^{-5}	-8.27	-12.2
t	172.56 GeV	53 800	3.2×10^{-3}	-2.89	-6.8
d	4.70 MeV	53 800	8.7×10^{-8}	-14.13	-18.0
s	93.5 MeV	53 800	1.7×10^{-6}	-11.02	-14.9
b	4.183 GeV	53 800	7.8×10^{-5}	-7.00	-10.9

($d\ln m/d\ln c=\frac{1}{2}\,d\ln\zeta(c)^2/d\ln c$ for the singlet zero mode with the doublet fixed; the doublet localisation enters every mass of its sector with -5.45 (leptons) and -2.89 (quarks); $d\ln m/d\ln f=-1+\frac{1}{2}[d\ln\zeta(c_L)^2/d\ln c_L+d\ln\zeta(-c_R)^2/d\ln(-c_R)]$, since both localisations $c=f\mu L$ are proportional to f and $m=e/(fL)$.) Proper spreads, uncertainty estimates, tip densities and uncertainty products of the bosons and of dirac16complex:

state	Δs (m)	$m_H=\hbar c/(2\Delta s)$ (GeV)	ϱ(tip)	$\langle p^2\rangle\Delta\xi^2$
photon, gluons, graviton; $W^\pm$; Z	1.40×10^{-21}	70 560	1.1×10^{-3}	0.497
first even Kaluza-Klein mode; first odd mode	6.09; 5.55×10^{-21}	16 204; 17 771	0.655; 0.433	4.59; 1.89
Ψ fold band; Ψ dark matter	4.31; 5.09×10^{-21}	22 903; 19 395	0.489; 0.772	0.097; 0.847

and the decomposition of the two Ψ states (the bulk term is $c_\psi^2=(f_*mL)^2=0.418$):

state	$\langle\lvert\chi'\rvert^2\rangle$	bulk c_ψ^2	boundary term	$(\varepsilon fL)^2$	m/m_H
Ψ fold band (both halves)	0.418	0.418	-0.836	0	0
Ψ dark matter (both halves)	2.608	0.418	0.998	4.024	2.29

Three classes follow.

- **Massless although localised.** The photon, the gluons and the graviton are localised at the fold ($\Delta s = 1.40 \times 10^{-21}$ m, $m_H = 70.6$ TeV, $\langle p^2 \rangle \Delta\xi^2 = 0.497$), yet exactly massless: the fold term cancels the hidden kinetic energy and the warp. The cancellation is exact because the flat profile is an eigenfunction with $\Lambda = 0$, protected by four-dimensional gauge invariance and general covariance. The fold bands of dirac16complex are massless for the analogous chiral reason ($\chi = (e^{f\int M dy}, 0)$ is a zero mode for every mass profile), and so is every fermion zero mode before the tip mixes it with its partner.
- **Heavy: the uncertainty estimate lies below the mass.** Every x_0 excitation of the gauge fields and the graviton lies above the gap of their operator: for every tip coupling β from 0 to 10^6 exactly one even level lies below $9\eta^2/4$ (the zero mode, lifted only by the tip) and no odd level does (the gap theorem, DERIVED), so every odd mode and every excitation of these fields has $m \geq 3H/(2f_*) = 99.5$ TeV. The Dirac operator of Ψ has no such warp gap; its excitations start at the lowest odd level, $3.10m = 44.35$ TeV (the dark matter). For the computed states m_H is exceeded by factors 2.3 to 7.5 (dark matter 2.29, first odd 6.47, first even 7.49): the author's Heisenberg estimate lies below these masses (observed for these states, not derived for all), and the dark-matter mass is an x_0 mass in this sense (its hidden wave number p carries $p^2/\varepsilon^2 = 0.896$ of ε^2).
- **Light: the mass comes from the tip, not from localisation.** For $W^\pm$, Z and the twelve fermions the x_0 dependence that costs energy sits at the tip, where the probability is small: $m/m_H = 1.1 \times 10^{-3}$ ($W^\pm$), 1.3×10^{-3} (Z), 3.2×10^{-3} (the top quark, a factor 312 below the estimate) down to 1.1×10^{-17} (ν_1). For the fermions a sharper localisation gives a LIGHTER particle, $d\ln m/d\ln c < 0$ for every zero mode (from -2.89 for the quark doublet and the top singlet to -34.55 for the ν_1 singlet): the reverse of the Heisenberg intuition. The lightest particles of all, the neutrinos, have the most sharply localised singlet partners ($\Delta s \approx 2 \times 10^{-22}$ m, $m_H \approx 0.5$ PeV).

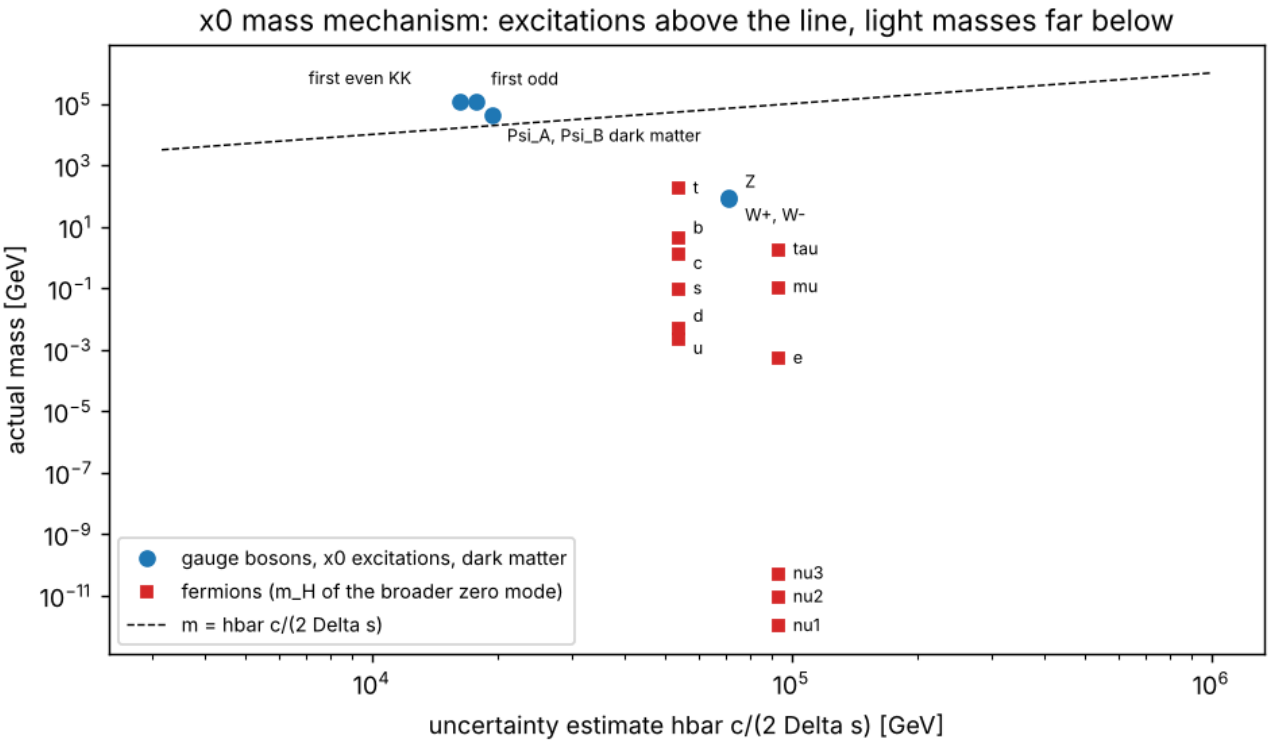


Figure 1: The x_0 mass mechanism: actual masses against the uncertainty estimate $m_H = \hbar c/(2\Delta s)$ (circles: bosons and dirac16complex; squares: the twelve fermions, with m_H of their broader zero mode); the excitations and the dark matter lie above the line, the light particles far below.

2.4 Proper versus coordinate localisation (DERIVED)

The photon zero mode, localised at the fold in proper length, has 61.9 % of its probability within $\pi/8$ of the tips in the coordinate $z = 6Hx_0$: localisation in the coordinate x_0 is chart dependent, because $ds_{hidden} = f \cot z \, dx_0$

vanishes at the fold and diverges at the tips. A profile confined to a small proper distance from the fold ($\Delta\xi = 0.157$, i.e. $|y| \lesssim 0.47/H$, $\sin z \gtrsim e^{-2.8}$) therefore extends over almost the whole coordinate range of z, and its probability per unit z is largest near the tips. Only the proper hidden length enters the masses; a statement "Δx_0 small" must therefore be read as "Δs small".

2.5 The neutrinos: masses, mixing and flavour conversion as x_0 effects

The three neutrino masses are x_0 eigenvalues of the same kind as the charged-fermion masses: Dirac masses produced at the tip by the overlap of the lepton doublet zero mode ($c_L = 5.9529$) with three right-handed singlet zero modes localised very sharply at the fold ($c_R = -35.05$, -32.85, -31.07). The right-handed neutrinos are gauge singlets; they are never in thermal equilibrium (their only interaction is the tip mixing, whose rate is suppressed by $(m_\nu/T)^2$; STANDARD), so they add no radiation. With the doublet shared, the charged current between the mass states ν_j and ℓ_i carries $\sum_\alpha (\kappa_\nu)^*_{\alpha j}(\kappa_e)_{\alpha i}$, so the PMNS matrix is $U = \kappa_e^\dagger \kappa_\nu = \kappa_\nu$ ($\kappa_e = 1$); its angles and phase are inputs of UM-2 (NuFIT 6.0), not predictions.

Given the masses and U, flavour conversion follows from the four-dimensional effective theory: a neutrino produced with flavour α is $|\nu_\alpha\rangle = \sum_i U^*_{\alpha i}|\nu_i\rangle$ and its flavour amplitudes $\psi = (\psi_e, \psi_\mu, \psi_\tau)$ evolve along the path with

$$i\,\frac{d\psi}{dx_4} = \Big[\frac{U\,\mathrm{diag}(m_1^2, m_2^2, m_3^2)\,U^\dagger}{2E} + \mathrm{diag}(V_{CC}, 0, 0)\Big]\psi, \qquad V_{CC} = \sqrt{2}G_F n_e$$

(E the energy, n_e the electron density; the neutral-current potential is common to the three flavours and drops out). The oscillation phases $\Delta m^2_{ij} L/(4E)$ are differences of squared x_0 eigenvalues, so oscillation experiments measure the x_0 spectrum directly. They are the quantities of UM-2 most sensitive to the block scale (DERIVED, from $d\ln(m_j^2 - m_i^2)/d\ln f = 2(m_j^2 s_j - m_i^2 s_i)/(m_j^2 - m_i^2)$ with $s_i = d\ln m_i/d\ln f$):

$$\frac{d\ln\Delta m^2_{21}}{d\ln f} = -77.5, \qquad \frac{d\ln\Delta m^2_{31}}{d\ln f} = -74.0, \qquad \frac{d\ln\Delta m^2_{32}}{d\ln f} = -73.9.$$

The light neutrino states also mix with the Kaluza-Klein excitations of the lepton fields, which makes the 3×3 PMNS matrix non-unitary at relative order $(m_i/m_{KK})^2 \le 1.7\times 10^{-31}$ (ESTIMATE; for the top quark 2.0×10^{-6}): unobservable. The flavour conversion of UM-2 is therefore that of the three-flavour Standard Model with Dirac neutrinos, which UM-2 computes for the vacuum, the Earth, the Sun, the early Universe and supernovae.

Dirac or Majorana (DERIVED, version 3). The x_0 mechanism explains the small Dirac masses by the localisation of the right-handed singlets with tip couplings of order one. Without right-handed neutrinos the masses would be Majorana masses from the gauge-invariant tip operator $-\frac{1}{2}\sum_{ij} m_{ij}\overline{N_i^c}N_j + \text{h.c.}$, $N_i = (\Sigma^\dagger L_i)_1$ (neutral, since $\Sigma \to U_L \Sigma U_Y^\dagger$), with

$$m_\nu = \kappa_M \frac{\zeta(c_L)^2}{fL}, \qquad \zeta(c_L)^2 = 8.04\times 10^{-5}, \qquad \frac{1}{fL} = 22.1\ \text{TeV},$$

which needs $\kappa_M = 5.6\times 10^{-13}$, 4.9×10^{-12}, 2.8×10^{-11}: a hierarchy of tip couplings that the mechanism does not explain. The x_0 mechanism therefore favours Dirac neutrinos. The Majorana case would show up in neutrinoless double-beta decay with $m_{\beta\beta} = |\sum_i U_{ei}^2 m_i e^{i\alpha_i}|$ between 0.84 and 4.41 meV (α_i the Majorana phases) and in a twice larger capture rate of the non-relativistic CNB; the flavour conversion is the same.

2.6 The mass mechanism, the block and the deflating extra times

Every mass and coupling depends on the block scale f (DERIVED):

quantity X	$d\ln X/d\ln f$
Kaluza-Klein masses, the gap $3H/(2f)$	-1
m_W (and m_Z)	-0.501
m_{ν_1}, m_{ν_2}, m_{ν_3}	-41.0, -38.8, -37.0
m_e, m_μ, m_τ	-20.5, -14.9, -11.9

quantity X	$d\ln X/d\ln f$
m_u, m_c, m_t	-18.8, -12.2, -6.8
m_d, m_s, m_b	-18.0, -14.9, -10.9
M_{DM}	-0.772
α and the gauge couplings ($g^2=\hat g^2/V_\perp\propto f^{-4}$)	-4
G_N ($8\pi G_N=\kappa/(G(k)V_\perp)$: $-4+2k_*G'(k_*)/G(k_*)$)	$+2.05$

The author's metric lets the extra times x_5, x_6, x_7 (and, in the corrected metric, x_0 with them) deflate exponentially. In UM-2 they do so in the primordial era: the full nonlinear equations take a large block into the stabilised one at rates up to $|h| = 2.63H$ (in the stored runs), deflating by up to $f(k_c)/f_* = 2.917$, after which the block oscillates about f_* (the radion) and stays frozen. A continuing deflation today is excluded (ESTIMATE, bounds FROM MEMORY): since $d\ln X/dx_4 = (d\ln X/d\ln f)\,h$, the atomic-clock bound $|d\ln\alpha/dx_4| < 10^{-17}$/yr ($x_4$ the time) and $d\ln\alpha/d\ln f = -4$ require $|h| < 2.5\times10^{-18}$/yr $= 3.6\times10^{-8}H_0$ ($H_0 = 6.90\times10^{-11}$/yr); lunar ranging ($|d\ln G_N/dx_4| < 10^{-13}$/yr, $d\ln G_N/d\ln f = +2.05$) requires $|h| < 4.9\times10^{-14}$/yr $= 7.1\times10^{-4}H_0$; a block deflating at the Hubble rate ($|h| = H_0$) would change the electron mass by $20.5H_0 = 1.4\times10^{-9}$ per year. The extra times are irrelevant at low energies not because they keep deflating but because they are small (the proper half cell $f_*L = 3f_*/H = 8.9\times10^{-21}$ m, Kaluza-Klein masses above 100 TeV) and frozen. The price is the stabiliser (ASSUMED, origin OPEN) and the radion left over from the deflation (ESTIMATE: a Planck-suppressed decay $\Gamma = C_r\, m_r^3/(4\pi\bar M_{Pl}^2)$, with the reduced Planck mass $\bar M_{Pl} = 2.435\times10^{18}$ GeV and an order-one coupling factor C_r, gives the lifetime $0.43\,\mathrm{s}/C_r$ and the reheating temperature $T_{RH} = (90/(\pi^2 g_*))^{1/4}\sqrt{\Gamma\bar M_{Pl}} = 1.85\sqrt{C_r}$ MeV, $g_* = 10.75$, below the 4.7 MeV that nucleosynthesis needs unless $C_r > 6.5$: a moduli problem, OPEN).

2.7 The nucleon mass and the block (version 3)

Most of the visible mass is nucleon mass, i.e. QCD binding energy, which is not an x_0 eigenvalue of a field but follows from the strong coupling by dimensional transmutation. In UM-2 the coupling is fixed at the Kaluza-Klein scale, $g_s^2 = \hat g_s^2/V_\perp \propto f^{-4}$, so the QCD scale is an x_0 quantity too (DERIVED, ESTIMATE at one loop): with $1/\alpha_s(m_{KK}) = 1/\alpha_s(m_Z) + \frac{7}{2\pi}\ln(m_{KK}/m_Z)$, $\alpha_s(m_{KK}) = 0.0606$, $\Lambda^{(6)} = m_{KK}e^{-2\pi/(7\alpha_s(m_{KK}))}$, the matching $\Lambda^{(3)} = \Lambda^{(6)\,21/27}(m_c m_b m_t)^{2/27}$ and the sigma terms $\sigma_{\pi N} = 45$ MeV, $\sigma_s = 40$ MeV,

$$\frac{d\ln\Lambda^{(6)}}{d\ln f} = -1 - \frac{8\pi}{7\alpha_s(m_{KK})} = -60.2, \qquad \frac{d\ln\Lambda^{(3)}}{d\ln f} = \frac{21}{27}\frac{d\ln\Lambda^{(6)}}{d\ln f} + \frac{2}{27}\sum_{q=c,b,t}\frac{d\ln m_q}{d\ln f} = -49.0,$$

$$\frac{d\ln m_p}{d\ln f} = \Big(1 - \frac{\sigma_{\pi N}+\sigma_s}{m_p}\Big)\frac{d\ln\Lambda^{(3)}}{d\ln f} + \frac{\sigma_{\pi N}}{2m_p}\Big(\frac{d\ln m_u}{d\ln f} + \frac{d\ln m_d}{d\ln f}\Big) + \frac{\sigma_s}{m_p}\frac{d\ln m_s}{d\ln f} = -46.1:$$

the nucleon mass is the most block-sensitive mass of UM-2, more than the lightest neutrino (-41.0); a block that drifted today would change the proton-to-electron mass ratio by $(46.1-20.5)|h|$ per unit time. In the coupled solution the hidden pressures of all species shift the block by at most $\delta\ln f = 5.6\times10^{-45}$ (at the dark-matter freeze-out; 10^{-62} in the Earth), so the masses are fixed to that accuracy.

The Kaluza-Klein gravitons couple to a source through the overlap of its x_0 profile with theirs; their profiles $u_n = e^{-a\xi}(\cos n\pi\xi + \frac{a}{n\pi}\sin n\pi\xi)$ ($a = 3\eta/2$) are orthogonal to the flat profile of the gauge zero modes in the measure $e^{3\eta\xi}$, so the gluon field energy that makes up most of the nucleon mass excites none of them (DERIVED to 2×10^{-10}): gravity at short distances feels the localised quark and lepton zero modes only, with Yukawa strengths 0.09 and 0.02 for the first mode and a range of 1.6×10^{-21} m.

2.8 Assessment of the mechanism

The author's mechanism works in UM-2 in a precise sense: all masses are x_0 eigenvalues, the period $\pi/(6H)$ is built into every wave function, and the massless particles are exactly those protected by symmetry. The uncertainty relation, however, is the reason for the masses of the heavy states only. The light masses need two further ingredients that are not consequences of the uncertainty relation: the tip of the cell (where electroweak symmetry is broken by Σ) and exponentially localised chiral zero modes, whose overlap at the tip ($\zeta(c)^2 \simeq 2c\,e^{-2c}$) produces the hierarchy

of the twelve fermion masses, fourteen orders of magnitude from ν_1 to the top quark, from bulk masses between $5.2H$ and $54.2H$ in magnitude. The mixing matrices are tip matrices and are not explained by the mechanism. The scales are not free: once the Kaluza-Klein scale is fitted ($m_{KK} = 121$ TeV from ρ_0), the dark-matter mass (44 TeV) and the radion (48 TeV) follow. Version 3 adds three observations: the nucleon mass, the bulk of the visible mass, depends on the block through the QCD scale more strongly than any x_0 eigenvalue (-46); the mechanism favours Dirac over Majorana neutrinos; and in the coupled solution every back-reaction on the masses (Standard-Model mean fields on dirac16complex, dirac16complex on the gauge couplings, all species on the block) stays below 1.6×10^{-6}, so the x_0 spectra of the static block are the masses of the solution.

3 dirac16complex as dark matter

3.1 The candidate

Ψ splits into two four-dimensionally Lorentz-invariant halves, Ψ_A ($B = -1$) and Ψ_B ($B = +1$), $B = -iC\gamma^{x_4}$. In the good sector every term of UM-2 commutes with B (DERIVED with the author's matrices: B commutes with C, with γ^a and $C\gamma^a$ for $a = x_0, \ldots, x_4$, anticommutes with them for $a = x_5, x_6, x_7$, and $\gamma^M \Omega_M = \frac{3}{2}\frac{H}{f}\gamma^{x_0} + (\frac{\dot{b}_4}{b_4} - \frac{\dot{a}_4}{2})\gamma^{x_4}$ contains γ^{x_0} and γ^{x_4} only), so N_A and N_B are separately conserved. The chiral tip projector $P_+ = (1 + \Gamma_4)/2 = (1 - B\gamma^{x_0})/2$ ($\Gamma_4 = -B\gamma^{x_0}$, DERIVED) annihilates every Ψ_B orbital at the tip, while a Ψ_A orbital has a non-zero chiral tip charge density (0.257 for the lightest odd state); UM-2 contains no tip operator (the chiral tip Z portal of version 1 is removed), so neither half is reached at the tip. The dark-matter candidate is the lightest odd state of each half:

- two Dirac fermions of mass $M_{DM} = \sqrt{m^2 + p^2} = 3.102204\, m = 44.35$ TeV ($\tan(p\, fL) = -p/m$, $m = H$), degenerate (the same orbital in all eight blocks);
- cold: $w = p_3/\rho = 0$ for quanta at rest (Kohn-Sham, DERIVED with the equations of the section on dark energy below; 2.8×10^{-5} with the benchmark self-coupling);
- electrically neutral (gauge singlets), with no Z coupling;
- stable: N_A and N_B are conserved, and the fold parity of each half is exact (the bulk is symmetric under the fold reflection, and UM-2 has no oriented tip operator that could connect an odd state to the even fold band);
- its mass is an x_0 mass: $p^2/\varepsilon^2 = 0.896$, $m/m_H = 2.29$, and its hidden pressure $p_0/\rho = 0.772 = -d\ln\varepsilon_{DM}/d\ln f$ (two independent computations agree to 10^{-9}).

3.2 Relic abundance

Through the gauge-kinetic portal $S_{I12} = -\int \sqrt{|g|}\, \bar{\Psi}\Psi(c_G G^2 + c_W W^2 + c_B B^2)/\Lambda_\star^3$ the four-dimensional operator is $G_i \bar{\chi}\chi F^i_{\mu\nu}F^{i\mu\nu}$ with $G_i = \mathcal{O}\, c_i g_i^2/\Lambda_\star^3$, $\mathcal{O} = \partial\varepsilon_{DM}/\partial m = 0.7072$. The annihilation is p-wave: $\sigma v = G^2 s(s - 4M^2)/(4\pi) \to G^2 M^4 v^2/\pi$ per abelian channel, $(M^4 v^2/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$ in total. Since $\bar{\Psi}\Psi = \bar{\Psi}_A\Psi_A + \bar{\Psi}_B\Psi_B$ does not mix the halves, each species freezes out separately and supplies half of $\Omega_c h^2 = 0.120$. The thermal relic (standard freeze-out in a radiation-dominated Universe, Dirac fermion, $g_* = 105.75$ at $T_f \approx 1.5$ TeV: the Standard Model of three generations without a Higgs particle; the right-handed neutrinos and the fold bands are never thermalised) requires (DERIVED)

$$x_f = \ln\left[0.038\, c(c+2)\, g\, M_{Pl} M \langle\sigma v\rangle / \sqrt{g_* x_f}\right] = 30.38, \qquad \Omega h^2 = \frac{2 \times 1.07 \times 10^9\ \mathrm{GeV}^{-1}\, x_f}{\sqrt{g_*}\, M_{Pl}\, 3b/x_f} = 0.060 \text{ per species},$$

($\langle\sigma v\rangle = 6b/x$, $b = (M_{DM}^4/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$, $c = 1/2$, $g = 2$, $M_{Pl} = 1.22091 \times 10^{19}$ GeV), i.e.

$$\langle\sigma v\rangle(x_f) = 2.01 \times 10^{-25}\ \mathrm{cm}^3/\mathrm{s} \text{ per species}, \qquad G \equiv (8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = (15.5\ \mathrm{TeV})^{-3}$$

(a single species carrying all the dark matter would need $(17.6\ \mathrm{TeV})^{-3}$ with the same g_*).

Unitarity (DERIVED). The operator $\bar{\chi}\chi F_{\mu\nu}F^{\mu\nu}$ annihilates a fermion pair only from the state with $J = 0$ and positive parity, ${}^3P_0$. Partial-wave unitarity bounds the inelastic cross-section from one $J = 0$ state by π/p^2; averaged over the four spin states of an unpolarised Dirac pair, of which one is ${}^3P_0$,

$$\sigma \le \frac{\pi}{4p^2}, \qquad p = \frac{M_{DM} v}{2}\ (v \ll 1), \qquad \sigma v \le \frac{\pi}{M_{DM}^2 v},$$

$$\left\langle \frac{\pi}{M_{DM}^2 v} \right\rangle = \frac{\sqrt{\pi x}}{M_{DM}^2} = 5.79\times10^{-26}\ \mathrm{cm}^3/\mathrm{s} \text{ at } x_f = 30.38,$$

the thermal average over the relative velocity with $f(v) \propto v^2 e^{-xv^2/4}$ (checked by numerical averaging). The freeze-out requirement exceeds the bound 3.47 times, and the effective cross-section $G^2 M_{DM}^4 v^2/\pi$ at the relic strength exceeds it for $v > v_c = (\pi/(GM_{DM}^3))^{2/3} = 0.263$, i.e. for 79 % of the pairs at freeze-out. Version 1 compared the requirement with $(4\pi/M_{DM}^2)\sqrt{x_f/\pi} = 2.32\times10^{-25}$ cm^3/s, the bound of a single helicity channel without the spin average and the restriction to ${}^3P_0$, four times too large. The gauge-kinetic portal I12 therefore cannot produce the relic abundance within perturbative unitarity (version 2).

The relic in version 3 (DERIVED). The Boltzmann equation of one Dirac species ($Y = (n_\chi + n_{\bar\chi})/s$, $x = M/T$),

$$\frac{dY}{dx} = -\frac{(1+\delta)s\langle\sigma v\rangle}{2xH}(Y^2 - Y_{eq}^2), \qquad \delta = \frac{1}{3}\frac{d\ln g_s}{d\ln T}, \qquad \langle\sigma v\rangle = \frac{\int_{4M^2}^\infty \sigma(s)(s-4M^2)\sqrt{s}\,K_1(\sqrt{s}/T)\,ds}{8M^4 T K_2(x)^2},$$

with $Y_{eq} = 4M^2 T K_2(x)/(2\pi^2 s)$, $H = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{Pl}$, $s = \frac{2\pi^2}{45}g_s T^3$, and the degrees of freedom $g_\rho(T)$, $g_s(T)$ of the particle content of UM-2 with its x_0 masses, confirms the conclusion and quantifies it: the effective I12 would need $(15.86\ \mathrm{TeV})^{-3}$, crossing the ${}^3P_0$ bound at $v = 0.274$; a cross-section at that bound at every energy, the largest annihilation from ${}^3P_0$, leaves $\Omega h^2 = 0.058$ per species, so I12 could produce the relic only as a strongly coupled channel at the unitarity limit, outside the validity of the effective description. Version 3 therefore keeps I12 at the unitarity-safe benchmark $G = \pi/(2M^3) = (38.2\ \mathrm{TeV})^{-3}$ (alone: $\Omega h^2 = 8.0$ per species) and adds the INVENTED portal

$$S_{I13} = \int d^8x\sqrt{|g|}\,\frac{c_A}{\Lambda_\star^6}(i\bar\Psi\gamma^M P_b\Psi)J_M^B, \qquad P_b = \frac{1-\gamma^{x_0}}{2}, \qquad J_M^B = \frac{1}{3}\sum_{i=1}^{3}\sum_{\psi\in\{Q_i,u_{Ri},d_{Ri}\}} i\bar\psi\gamma_M\psi,$$

whose x_0-chiral current is Hermitian, a four-dimensional vector, diagonal in the halves, and zero on the fold bands ($\gamma^{x_0} = \sigma_3$ in the block form and the fold bands have no lower component). On the zero modes it couples the dark matter to each quark Weyl component f with $G_f = G_A\frac{1}{3}\tilde I_f^{(b)}$, $\tilde I_f^{(b)} = \int_{-L}^0 b^2\varrho_f e^{-3Hy}dy$ (b the lower component of the dark-matter orbital, ϱ_f the zero-mode profile), and the dark matter annihilates in the s-wave,

$$\sigma = \sum_f \frac{N_c G_f^2\, s\,(1+2M^2/s)}{24\pi\beta} \;\le\; \frac{3\pi}{4p^2} \qquad (\beta = \sqrt{1-4M^2/s},\ J=1).$$

Each species then freezes out with $\Omega h^2 = 0.060$ for $G_A = 1.278\times10^{-9}\ \mathrm{GeV}^{-2}$ ($1.1\times10^{-4}G_F$; $x_f = 31.2$), the cross-section reaching its bound only above $v = 0.51$ (19 % of the freeze-out rate): a perturbative thermal relic. The current must be the baryon-number current: a $B-L$ current would bring the three right-handed neutrinos into equilibrium through $\chi\bar\chi \leftrightarrow \nu_R\bar\nu_R$ until $T = 1.8$ TeV and add $\Delta N_{eff} = 3(10.75/g_s)^{4/3}F_h = 0.144$, 2.5 standard deviations above ACT DR6. The relic value assumes that the radion oscillation left over from the deflation, which redshifts like dust, does not dominate and dilute the Universe after the freeze-out (the moduli problem).

3.3 Detection

- Direct (ESTIMATE): even if the whole coupling were in gluons ($G_g = G/\sqrt{8}$), the trace anomaly $\langle N|G^\alpha_{\mu\nu}G^{\alpha\mu\nu}|N\rangle = -\frac{8\pi}{9\alpha_s}m_N f_{TG}$ gives the nucleon coupling and cross-section

$$f_N = G_g\,\frac{8\pi}{9\alpha_s}\,m_N f_{TG}, \qquad \sigma_{SI} = \frac{\mu_N^2 f_N^2}{\pi} = 6.3\times10^{-53}\ \mathrm{cm}^2$$

  ($f_{TG} = 0.92$, $\alpha_s = 0.3$ at a low scale, $m_N = 0.939$ GeV, μ_N the dark-matter-nucleon reduced mass; FROM MEMORY) for each half (the two halves have the same cross-section and together the full local density, so the limit applies as for one species), eight orders below the LZ limit extrapolated linearly in the mass from 2.2×10^{-48} cm^2 at 40 GeV to 44 TeV (2.4×10^{-45} cm^2): the limit would allow a gluon coupling whose square is 3.9×10^7 times the relic requirement. The electroweak channels have no tree-level nucleon coupling.
- Direct, version 3 (DERIVED, ESTIMATE for the limit): I13 (and I11) couple the dark matter coherently to the quark number. With $G_{u,d} = G_A\frac{1}{3}\frac{1}{2}(\tilde I_Q^{(b)} + \tilde I_{u_R,d_R}^{(b)}) + G_V\frac{1}{3}\frac{1}{2}(\tilde I_Q + \tilde I_{u_R,d_R})$,

$$f_p = 2G_u + G_d, \qquad f_n = G_u + 2G_d, \qquad \sigma_n = \frac{\mu_N^2}{\pi}\Big(\frac{Zf_p + (A-Z)f_n}{A}\Big)^2 = 4.0\times10^{-46}\ \mathrm{cm}^2$$

for xenon ($Z = 54$, $A = 131$; $f_p = 1.915 \times 10^{-9}$, $f_n = 1.917 \times 10^{-9}$ GeV^{-2}), 0.16 of the extrapolated LZ limit: the dark matter of UM-2 is testable by the next generation of xenon detectors.

- Indirect: the I12 annihilation is p-wave, $\propto v^2 \sim 10^{-6}$ in galaxies today; the I13 annihilation is s-wave and continues today with $\sigma v = \sum_f N_c G_f^2 M^2/(2\pi) = 9.8 \times 10^{-26}$ cm^3/s into quark pairs, the thermal value for a heavy relic.
- Colliders: 44 TeV is beyond the reach of present and planned colliders.

3.4 The massless quanta are not dark matter

The fold bands of both halves are massless four-dimensional fermions ($\varepsilon = e^{-a_4}|k|$); a gas of them has $w = 1/3$ exactly (Kohn-Sham, DERIVED). In UM-2 they are never thermalised: there is no tip operator, the scalar density of a chiral zero mode vanishes identically (so I12 does not touch them), I13 vanishes on them, and the vector portal I11 is weak. In version 3 I11 has its benchmark strength (with the $B - L$ charges q_f and the fold-band overlaps $\tilde{I}_f$): the plasma produces fold-band pairs by freeze-in at the rate $\gamma = \frac{4}{\pi^5} N_\chi G_V^2 T^8 \sum_f N_c (q_f \tilde{I}_f)^2$ ($N_\chi = 2$; each pair carries $8T$), which from $T_{max} = 10$ TeV adds $\Delta N_{eff} = 0.001$ for $G_V = 4.38 \times 10^{-17}$ GeV^{-2} without ever reaching equilibrium: a trace of dark radiation, not dark matter, and $N_{eff} = 3.0442$ instead of 3.0432. Had one half's fold band (two Weyl species) been thermalised, it would add (DERIVED)

$$\Delta N_{eff} = n_W \Big(\frac{10.75}{g_s}\Big)^{4/3} F_h, \qquad n_W = 2, \qquad F_h = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3} \Big/ (1 + h_\nu)\Big]^4 = 1.0095,$$

(g_s the entropy degrees of freedom at its decoupling; F_h corrects for the heating of the neutrinos by e^+e^- annihilation that an earlier-decoupled species does not receive, with the computed neutrino-to-photon temperature ratio $T_\nu/T_\gamma = 0.71664$ and the comoving heating $h_\nu = 0.00165$ of one common neutrino temperature), i.e. 0.0958 for a decoupling above the electroweak scale ($g_s = 105.75$) and 0.1963 just above the QCD crossover ($g_s = 61.75$); at fixed Planck parameters these change the CMB temperature spectrum by up to 1.5 % and 3.1 % for $30 \le \ell \le 2500$, and put N_{eff} 2.1 and 2.9 standard deviations above the ACT DR6 value 2.86 ± 0.13. Whether the fold bands are produced gravitationally during the primordial deflation is OPEN.

3.5 Assessment

The association of dirac16complex with dark matter is valid in UM-2 version 3 under stated conditions: (i) the good sector (no extra-time momenta; outside it the halves mix and modes are tachyonic, OPEN); (ii) the exact fold parity of both halves, which UM-2 secures by containing no oriented tip operator ($P_+\Psi_B = 0$ at the tip even protects Ψ_B against any chiral tip current); (iii) fold bands that never thermalise (I11 weak, I12 and I13 blind to them, no dipole portal, no significant gravitational production); (iv) an annihilation channel that produces the relic within unitarity: in version 2 the gauge-kinetic portal I12 alone could not (its freeze-out requirement exceeds the $^3P_0$ bound 3.5 times); in version 3 the INVENTED s-wave portal I13 to the baryon-number current does, with $G_A = 1.28 \times 10^{-9}$ GeV^{-2}, and predicts a direct-detection signal at 0.16 of the present LZ limit; (v) no late dilution by the radion (the moduli problem). The candidate is falsifiable: a xenon exposure about six times that of LZ would test it. It is the heavy odd part of the field, not the field as a whole, that is dark matter; the field's massless fold bands are radiation ($\Delta N_{eff} = 0.001$).

4 dirac16complex as dark energy

4.1 No positive-energy state accelerates (DERIVED)

The many-body states of Ψ on the static block (units $H = 1$, $a_4 = 0$, $f = f_*$) are described by Kohn-Sham orbitals χ_o,

$$h_j \chi_o = \varepsilon_o \chi_o, \qquad h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + M_{eff}\sigma_2 + e^{-a_4}k\sigma_3\Big] + v, \qquad M_{eff} = m + \frac{15}{16}\lambda S + \frac{\langle \mathcal{F} \rangle}{\Lambda_\star^3}, \qquad v = -\frac{\lambda}{16}n + v_{I11},$$

$$n = \sum_o w g f_o n_o, \quad S = \sum_o w g f_o s_o, \quad n_o = P\chi_o^\dagger \chi_o, \quad s_o = P\chi_o^\dagger j\sigma_2 \chi_o, \quad t_o = P\chi_o^\dagger j\sigma_3\chi_o, \quad P = \frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

$$\rho = \sum_o w g f_o \varepsilon_o n_o - e_{int}, \qquad p_3 = \sum_o w g f_o \frac{e^{-a_4}|k|}{3} t_o + e_{int}, \qquad e_{int} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2,$$

with the fold conditions $\chi_2(0) = 0$ (even) or $\chi_1(0) = 0$ (odd) and the tip condition $\chi_2(-L) = 0$; $w = 1/2$ per fold parity, $g = 4r_3(|\vec{n}|^2)$ the degeneracy of the momentum shell $\vec{k} = \Delta k\,\vec{n}$ ($\Delta k = 0.25$, r_3 the number of representations as a sum of three squares), occupations f_o, $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 f_*^4 = 34.27$; λ is the contact self-coupling ($\lambda_B = 7 \times 10^{-5} H^{-6}$ the benchmark), $\langle \mathcal{F} \rangle / \Lambda_\star^3$ the mean field of I12 and v_{I11} that of I11 (both zero for the states below). Every computed state has $w = \int p_3 / \int \rho \geq 0$ (integrals $2\mathrm{Vol}_7 \int e^{3Hy}(\cdot)\,dy$): eight dark-matter quanta at rest 0 (2.8×10^{-5} with λ_B); the fold-band gas of 592 quanta 1/3 (0.3312 with $+\lambda_B$, 0.3354 with $-\lambda_B$).

4.2 The condensate runs away (DERIVED)

A homogeneous condensate $S = \bar{\Psi}\Psi \propto a^{-3}$ with $U = \frac{\lambda}{2} S^2$ has $\rho = mS + U$, $p = S\,dU/dS - U = \frac{\lambda}{2} S^2$, so

$$w(a) = \frac{X(a)}{1 + X(a)}, \qquad X(a) = \frac{\lambda S}{2m} = X_0 a^{-3}, \qquad X_0 = \frac{w_0}{1 - w_0}, \qquad w_a \equiv -\frac{dw}{da}\Big|_{a=1} = \frac{3X_0}{(1 + X_0)^2}.$$

$w < 0$ needs $\lambda < 0$, and then the energy density $\rho = mS(1 + X)$ crosses zero at $a = (-X_0)^{1/3}$:

w_0	-1.2	-1	-0.9	-0.861	-0.764	-0.5	-0.3
w_a (tangent)	-7.92	-6.00	-5.13	-4.81	-4.04	-2.25	-1.17
energy zero at z	0.224	0.260	0.283	0.293	0.322	0.442	0.630

The supernova fits of the author's private file, a constant $w = -0.764$ and $(w_0, w_a) = (-0.861, -0.60)$, are not reproduced: a condensate tuned to $w_0 = -0.861$ has $w_a = -4.81$ and negative energy density beyond $z = 0.29$.

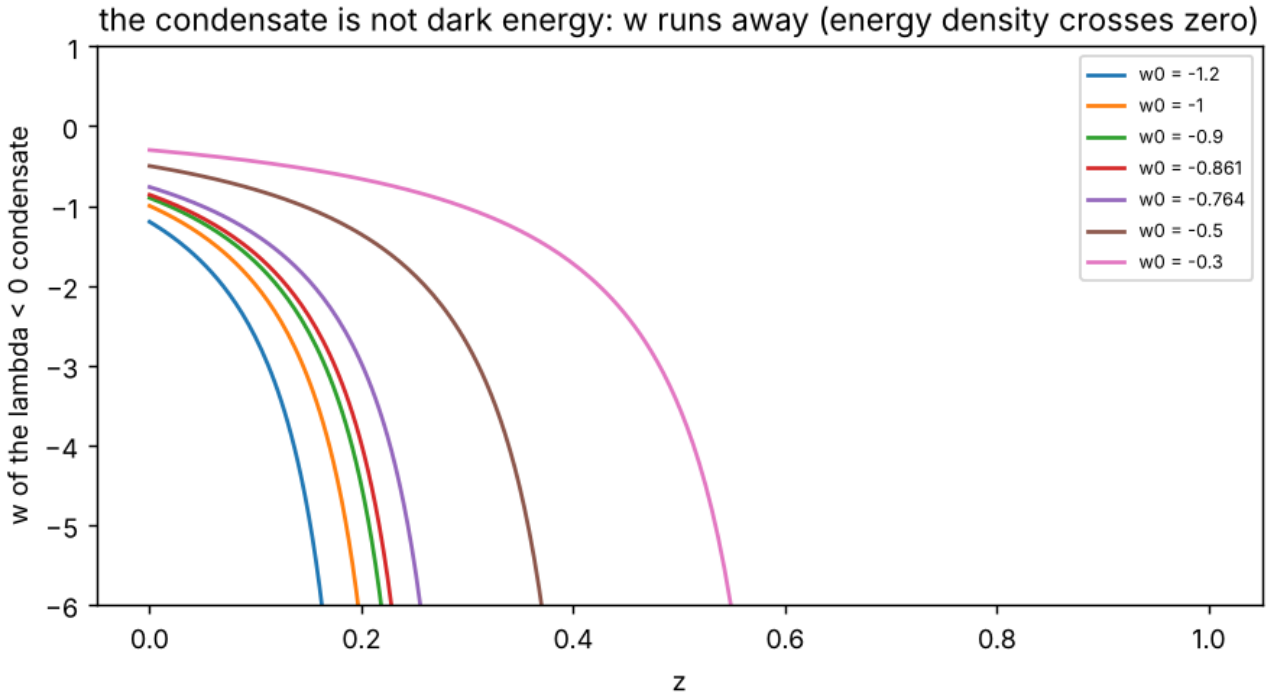


Figure 2: The equation of state of the $\lambda < 0$ condensate for several present values w_0: w runs away to $-\infty$ where the energy density crosses zero.

4.3 The vacuum energy and the tuning

A natural vacuum energy of the cell, $m_{KK}^4/(16\pi^2) = 1.4 \times 10^{18}$ GeV^4, exceeds the dark-energy density (2.5×10^{-47} GeV^4) by 5.5×10^{64}. In UM-2 the four-dimensional cosmological term is the residual of the stabiliser balance, $\Lambda_4 = [6H^2k(1-2w_2k)+\Lambda+\kappa\rho_b]/G(k)$, zero at the static point; the observed dark energy needs $\Lambda_4/H^2 \sim (H_0/H)^2 = 1.0 \times 10^{-92}$: the cosmological-constant problem, unsolved (OPEN). A slowly deflating block cannot provide a dynamical dark energy instead, because it would make α, G_N and the masses drift (the bound $|h| < 3.6 \times 10^{-8} H_0$ above), and the neutrino oscillation lengths most of all.

4.4 Assessment

The association of dirac16complex with dark energy is not supported in UM-2: its positive-energy states are matter or radiation, its only accelerating configuration is unstable and reaches negative energy density at $z \approx 0.2$ to 0.6, and the measured (w_0, w_a) are not reproduced. The dark energy of UM-2 is the tuned four-dimensional cosmological term Λ_4.

5 Conditions, open problems and conflicts

- **Conditions on which the conclusions rest** (ASSUMED): the stabiliser and its f^{-8} law; the good sector; the regular tip at $\eta = 3$; $m = H$; the tip sigma field, the tip matrices and the boundary assignments of the Standard-Model fields of three generations; the lightest neutrino mass $m_1 = 1$ meV and the normal ordering; the portals I11, I12 and I13 (INVENTED) with the benchmark strengths of I11 and I12 and the highest temperature $T_{max} = 10$ TeV (ASSUMED; the strength of I13 follows from the relic abundance).
- **Open**: the origin of the stabiliser; the extra-time modes; the closed time-like curves implied by compact extra times; the radion moduli problem; the block-shape mode driven by $p_0 - p_t$; the microscopic origin of the portals (I12 alone could give the relic only as a strongly coupled $^3P_0$ channel; version 3 obtains it through I13); the gravitational production of the fold bands; the origin of the tip matrices (the mixing angles); Dirac or Majorana neutrinos (the mechanism favours Dirac); the cosmological-constant problem.
- **Conflicts with data inherent in the order**: no Higgs boson (one is observed), and perturbative unitarity of W^+W^- scattering lost near 1.2 TeV. (The one-generation conflict of version 1, a Z width of 1.23 GeV, is removed: three generations give 2.485 GeV against the measured 2.4955 GeV.)